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Recovery of temperate and boreal forests after windthrow and the impacts of salvage logging. A quantitative review

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A R T I C L E I N F O

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A B S T R A C T

Wind is a natural disturbance factor in boreal and temperate forests having large ecological and economic consequences. We investigated recovery processes in forests severely damaged by windthrow, by conducting a systematic quantitative literature review of 34 case studies. We addressed three questions on forest ecosystem recovery: (1) Which forest regeneration processes dominate forest recovery after windthrow? (2) Which structures and processes enhance or impede forest recovery after windthrow? (3) Does salvage logging after windthrow influence forest recovery? Our analyses showed that the main focus of the reviewed studies was on post storm seedling regeneration and survival. Advanced regeneration from suppressed seedlings or saplings also played an important role in the studies, especially for shade tolerant species and in the boreal biome. Pits and mounds and coarse woody debris played an important role for the establishment of *Picea* sp., especially in the studies from the boreal biome, whereas in the temperate biome game interference and competition from ground vegetation seemed to be the most important influencing processes. Salvage logging mostly acted as a subsequent disturbance after the windthrow, pushing the forest ecosystem towards an earlier successional stage, hereby impeding recovery. In none of the studies however, did the windthrow nor the consequent salvaging halt recovery completely, confirming the resilience of forests as the dominant vegetation type in these biomes. Based on our results forest management implications are discussed.

1. Introduction

Temperate and boreal forests in the Northern Hemisphere are frequently damaged by storms, having large economic and ecological consequences (Gardiner et al., 2016). Focus of forest management in relation to windthrow is commonly aimed at improving single tree and stand resistance to wind. The use of robust deciduous species, early thinning, and shelter belts are frequently used to enhance forest stability (Gardiner et al., 2013). The periodical occurrence of extreme heavy wind events often happens several times during the life cycle of a forest stand (Senf and Seidl, 2018). The subsequent effects on the forest ecosystem and the economic return can never be fully prevented. It is therefore imperative that we study these effects to improve our management opportunities.

Planting and direct seeding after windthrow are common practice and have been intensively investigated, whereas structures and processes important for natural forest recovery are less frequently investigated. Observations from natural forests indicate less severe impacts by storms and faster recovery than in comparable managed forests (Mason, 2002; Wolf et al., 2004; Anyomi et al., 2016), indicating that forest management after windthrow may be amended by learning from natural forests (Jactel, 2017). Such observations may also increase our knowledge on passive restoration ('do nothing') after storm damage and what opportunities for cost-effective restoration and potential difficulties this offers to forest managers (Harmer and Morgan, 2009; Birch et al., 2010).

Natural recovery of forest ecosystems after windthrow relies on various forest regeneration mechanisms (Box 1). Tree species with strong anchorage may have a higher probability of surviving storms, and forests with survivors of such species are able to expand their canopies into wind induced canopy gaps through crown expansion (Coutts et al., 1999; Wolf et al., 2004). Some tree species have a

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Box 1
Clarification of concepts

Succession. Development of structures, processes, and interactions taking place in an ecosystem over time. Succession is *per se* natural and without intentional interventions and therefore open-ended.
Forest recovery. The sum of processes which lead to regaining of structures and attributes characterising forest ecosystems after a disturbance, such as presence of trees that form a crown cover, forest microclimate, and interactions of organisms typical of forest ecosystems. This definition opens for recovery leading to a 'new forest ecosystem' compared to the one present before disturbance.
Restoration. Intentional interventions in an ecosystem aiming at accelerating the recovery of a forest ecosystem to a reference or a natural trajectory. Passive restoration is the managerial decision to allow for open-ended succession aiming at the recovery of the forest ecosystem.
Forest regeneration. Recruitment or regrowth of trees that take over when conditions become favourable, commonly due to a disturbance of the crown canopy or death of neighbouring trees. Regeneration might be artificial (planting or direct sowing) or natural. In this paper we only consider natural regeneration after windthrow. We consider four regeneration processes:

- **Post storm regeneration.** Regeneration germinating from seed after canopy opening, here windthrow.
- **Survivors.** Surviving trees that were past the sapling stage at the time of the storm.
- **Advanced regeneration.** Suppressed seedlings or saplings that germinated before the storm and may be released if the storm opens up the canopy.
- **Vegetative regrowth.** Trees that resprout from stumps, roots or stems after sustaining windthrow.

Resilience. The capacity of a forest ecosystem to recover after a disturbance.
Resistance. The capacity of a forest ecosystem to buffer the effect of a disturbance and maintain structures and attributes.

capacity to resprout (Peterson and Pickett, 1995; Lang et al., 2009) or are present as advanced regeneration (Nagel et al., 2006; Waldron et al., 2013), and yet others have strong seed dispersal traits, enhancing potential colonisation after windthrow (Nakagawa et al., 2003). Forest recovery may be positively influenced by microsites such as pits and mounds from uprooted trees and fallen woody debris, offering favourable conditions for seed germination and protection from game (Samonil et al., 2010; Vodde et al., 2010; Vodde et al., 2011). Both trees that survived the windthrow and the microsites provided by deadwood, pits and mounds are legacies of the windthrow, which have been shown to increase forest resilience (Seidl et al., 2014). However, several processes might negatively influence these recovery processes. Inter and intra specific competition between trees and ground vegetation may also slow down or impair tree establishment (Davis et al., 1998; Provendier and Balandier, 2008). Interference from game, which might prefer foraging in open areas, may destroy newly established trees (Kuijper et al., 2009) or impair vegetative regrowth (Myking et al., 2011; White et al., 2014). Outbreaks of pests and diseases or unfavourable climatic conditions may also kill surviving trees (Khakimulina et al., 2016). These processes may influence forest recovery and the successional direction of the forest after windthrow (Peterson and Leach, 2008b).

Salvage logging after windthrow has been hypothesised to have cumulative detrimental effects to forest ecosystems, by further damaging the forest climate and homogenising the structural diversity (Kuuluvainen and Kalmari, 2003; Lindenmayer and Noss, 2006; Michalova et al., 2017). Rapid recovery of the canopy by survivors, advanced regeneration and by vegetative regrowth may be undermined by harvesting operations, leading to loss of forest microclimate and increased competition by grasses and herbs and selective browsing by game (Peterson and Leach, 2008a). The cumulated effect of windthrow and salvage logging (Paine et al., 1998), or the interactions between several disturbances, e.g. repeated windthrows (Buma and Wessman, 2011) may lower forest resilience (Seidl et al., 2014). It may potentially change the successional direction of the forest, and in severe cases push forest ecosystems towards irreversible thresholds (Peterson and Leach, 2008a). A consequence may be forest degradation and the creation of open landscapes for extended periods of time (Engelmark et al., 1998; Paine et al., 1998).

The research on natural forest recovery and effects of salvage logging seems to be relatively scarce and scattered in terms of time, biome, approach and research focus. To get an overview, we therefore found it useful to review the literature using a systematic quantitative review (Pickering and Byrne, 2014). The specific aims of our study were to investigate forest recovery with and without salvage logging after a windthrow event. To this end we formulated three key research questions:

1. Which forest regeneration processes dominate forest recovery after windthrow?
2. Which structures and processes enhance or impede forest recovery after windthrow?
3. Does salvage logging after windthrow influence forest recovery?

To answer question #1 we focused on studies that have investigated the role of post storm regeneration, surviving trees, advanced regeneration and vegetative regrowth. Considering question #2 we first focused on studies that investigate structures and processes that enhance forest recovery, for example soil disturbances enhancing germination of seeds, presence of coarse woody debris as microsite for tree establishment, and the presence of seed sources in close proximity. Secondly we focused on structures and processes that potentially impede forest recovery, for example effects of game interference, competing ground vegetation or outbreaks of pests and diseases. Finally for answering question #3 we focussed on studies which investigate how salvage logging after windthrow affects forest recovery. To structure our investigation sub-questions were formulated in relation to the three key research questions (Table 1).

We used peer reviewed studies as the source for the systematic quantitative literature review. The heterogeneity among investigated parameters in the relevant studies was too large and the number of relevant studies too small for a standard *meta*-analysis including statistical analyses. We applied this approach to investigate our research questions but also to identify areas well investigated and potential research needs. Finally, we discuss our results in relation to forest management to provide substance for discussion among scientists and practitioners about the use of natural recovery mechanisms as a viable alternative strategy of forest recovery after windthrow.

2. Materials and methods

We focused on the boreal and temperate forest biomes in the northern hemisphere. We found two clusters of case studies within this geographical area. Our investigation therefore included studies from Europe (to the Ural Mountains) and from the eastern USA and Canada (Fig. 1).

2.1. Case study search and selection criteria

Studies investigating forest recovery after windthrow with and without salvage logging and studies on how forest structures and

Table 1
Key research questions and related sub-questions.

Key research questions	Sub-questions
Which forest regeneration processes dominate forest recovery after windthrow?	1a) Does post storm regeneration contribute to forest recovery after windthrow? 1b) Do surviving trees contribute to forest recovery after windthrow? 1c) Does advanced regeneration contribute to forest recovery after windthrow? 1d) Does vegetative regrowth contribute to forest recovery after windthrow?
Which structures and processes enhance or impede forest recovery after windthrow?	2a) Does soil disturbance enhance tree establishment? 2b) Does coarse woody debris enhance tree establishment? 2c) Does seed source proximity enhance recovery? 2d) Does game interference retard recovery? 2e) Does competing ground vegetation retard forest recovery? 2f) Does leaving windthrown trees lead to pests and diseases?
Does salvage logging after windthrow influence forest recovery?	Does salvage logging 3a) prevent forest recovery, 3b) retard forest recovery, 3c) have no effect on forest recovery, 3d) change the successional pathway of forest recovery, 3e) increase the competition from ground vegetation, 3f) increase game interference to the recovering forest, 3e) enhance tree regeneration due to soil scarification?

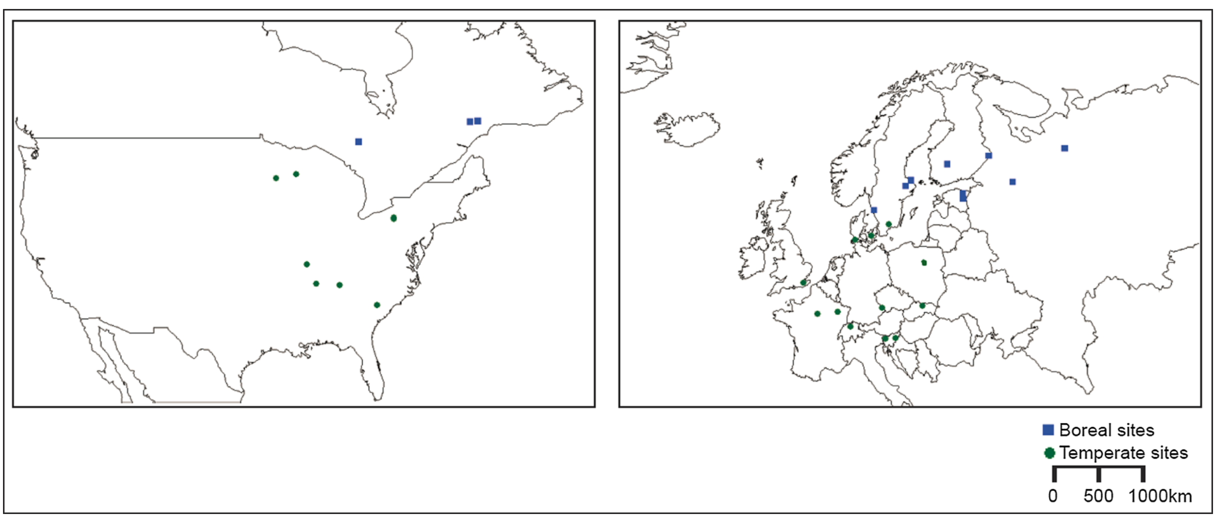

Fig. 1. Location of case studies included in the analyses.

processes influence forest recovery after windthrow were included. The case-studies were identified by searching web of science and google scholar with search keywords ‘forest’, ‘recovery’, ‘windthrow’, ‘blowdown’, ‘storm’, ‘wind’, ‘regeneration’, ‘succession’, ‘post-disturbance logging’, ‘sanitary logging’, ‘salvage logging’, ‘recruitment’, ‘establishment’ and ‘resilience’ in all possible combinations. Using all search keywords together gave no hits. We also used citations within read articles to look for further case studies. The few existing literature reviews on the topic were also used to identify case studies. All case studies were original investigations of recovery after windthrow and if investigated also following salvage logging after windthrow (Ulanova, 2000; Vodde et al., 2011). Approximately 60 studies were evaluated for relevance in relation to our purpose. Studies on planting of seedlings or direct seeding were not included nor were studies of forests damaged by forest fires or avalanches. While studies with a direct anthropogenic influence, except salvage logging were excluded from the dataset, we cannot avoid indirect effects on recovery from historic management or adjacent managed stands. These effects were not commonly mentioned in the studies and therefore they were not further regarded in this study. A few of the studies were conducted at the same site and by the same authors. These studies were only included once, except if they differed substantially in terms of their temporal scale and if additional structures or processes were investigated. Most of the case studies are about recovery after a discrete storm event. Some mention repeated windthrow or interacting disturbances, commonly not distinguishing between their effects, which does not allow us to analyse these effects separately. A total of 34 case studies finally met our criteria (Table 2).

2.2. Method selection

Statistical analyses of our data were attempted in the early stages of the research. For example, we tested which of the regeneration processes are most important, both with non-parametric tests and with a cumulative link mixed model. The low number of case studies, the non-paired observations and the heterogeneity among studies greatly weakens any statistical test. Therefore we have chosen the systematic quantitative review approach (Leverkus et al., 2015), in which we have included all details that our case studies show, to give an overview of forest recovery after windthrow. We did this as we found the gain of this more valuable than a weak statistical test.

2.3. Analyses of case studies

The case studies did not necessarily investigate all three key research questions. Most of the studies only investigated parts of the forest recovery process after windthrow, e.g. the role of pits and

Table 2
Case studies included in the analyses.

Site *	Biome	Main species	Year of storm	Year of assessment	Forest type	Investigated	Reference
Draved, Denmark	Temperate	*F. sylvatica, Q. robur*	1999	2000	Forest reserve	Free succession	Wolf et al., 2004
Suserup, Denmark	Temperate	*F. sylvatica, F. excelsior, U. glabra*	1999	2000–2001	Forest reserve	Free succession	Bigler and Wolf, 2007
Siggaboda, Sweden	Boreal	*P. abies, F. sylvatica*	2005	2004–2011	Forest reserve	Free succession	Bolte et al., 2010
Labra Park, Sweden	Boreal	*P. abies, Q. robur*	1969	1974–1984	Forest reserve	Salvage logging	Gotmark and Kiffer, 2014
Fiby, Sweden	Boreal	*P. abies*	1796	1984–1988	Forest reserve	Free succession	Leemans, 1991
Nuijakorpi, Finland	Boreal	*P. abies*	1950	2001	Forest reserve	Free succession	Kuuluvainen and Kalmari, 2003
Petkeljärvi, Finland	Boreal	*P. sylvestris*	1982	1994	Forest reserve	Free succession	Kuuluvainen and Juntunen, 1998
Tudu, Estonia	Boreal	*P. abies*	2001 + 2002	2005	Forest reserve	Free succession Salvage logging	Ilisson et al., 2007
Halliku, Estonia	Boreal	*P. abies*	2001 + 2002	2004–2007	Forest reserve	Free succession	Vodde et al., 2010
Tudu, Estonia	Boreal	*P. abies*	2001 + 2002	2004 + 2010	Forest reserve	Free succession	Vodde et al., 2015
Arkhangelsk, Russia	Boreal	*P. abies*	1990–2006	2009	Forest reserve	Free succession	Khakimulina et al., 2016
Vepssky, Russia	Boreal	*P. abies*	1983 + 1985	1971–2006	Forest reserve	Free succession	Shorohova et al., 2008
Kent & East Sussex, England	Temperate	*Mixed deciduous*	1987	2006	Managed forest	Free succession Salvage logging	Harmer and Morgan, 2009
Kent & East Sussex, England	Temperate	*Mixed deciduous*	1987	2004	Managed forest	Free succession Salvage logging	Harmer et al., 2004
Lorraine, France	Temperate	*F. sylvatica*	1999	2008	Managed forest	Free succession Salvage logging	Dodet et al., 2011
Tillaie, France	Temperate	*F. sylvatica*	1967	1981 + 1990	Forest reserve	Free succession	Pontailler et al., 1997
Jura & Central Plateau & Alps, Switzerland	Temperate	*P. abies, A. alba, F. sylvatica*	1990 + 1999	2009–2011	Managed forest	Free succession Salvage logging	Kramer et al., 2014
Tatra, Slovakia	Temperate	*P abies, L. decidua*	2004	2008	Managed forest	Free succession Salvage logging	Jonasova et al. (2010)
Slovenia	Temperate	*F. sylvatica, A. alba*	1993–2008	2012	Managed forest	Free succession Salvage logging	Fidej et al., 2016
Bavaria, Germany	Temperate	*P. abies, A. alba, F. sylvatica*	1983	1988 + 1993 + 1998	Forest reserve	Free succession Salvage logging	Fischer et al., 2002
Bavaria, Germany	Temperate	*P. abies, A. alba, F. sylvatica*	1983	1988–2008	Forest reserve	Free succession Salvage logging	Fischer and Fischer, 2012
Pecka, Slovakia	Temperate	*P. abies, A. alba*	1983	2004	Forest reserve	Free succession	Nagel et al., 2006
Allegheny Pennsylvania, USA	Temperate	*A. rubrum, Betula* sp.	1985	1989	Forest reserve	Free succession	Peterson and Pickett, 2000
Natchez Tennessee, USA	Temperate	*Quercus* sp.	1999	2001	Managed forest	Free succession Salvage logging	Peterson and Leach, 2008a
Shawnee Illinois, USA	Temperate	*Quercus* sp.	2009	2008 + 2010	Forest reserve	Free succession	Holzmueller et al., 2012
Cedar Creek Minnesota, USA	Temperate	*Quercus alba, Pinus* sp.	1983	1983–2008	Forest reserve	Free succession	Allen et al., 2012
North Shore Québec, Canada	Boreal	*P. mariana, A. balsamea*	2003–2006	2010	Managed forest	Free succession Salvage logging	Waldron et al., 2013
Flambeau Wisconsin, USA	Temperate	*T. canadendis, A. saccharum*	1977	2004	Managed forest	Free succession Salvage logging	Lang et al., 2009
Allegheny Pennsylvania, USA	Temperate	*Acer* sp., *F. grandifolia*	2003	2011	Managed forest	Free succession Salvage logging	Royo et al., 2016
Kapuskasing Ontario, Canada	Boreal	*P. tremuloides*	2011	2013	Managed forest	Free succession Salvage logging	Man et al., 2013
Natchez Tennessee, USA	Temperate	*Quercus* sp.	1999	2001	Managed forest	Free succession Salvage logging	Peterson and Leach, 2008b
Sipsey Alabama, USA	Temperate	*Q. alba*	2011	2013	Forest reserve	Free succession Salvage logging	White et al., 2014
North Shore Quebec, Canada	Boreal	*A, balsamea, P. mariana*	2011	2011	Managed forest	Free succession	Girard et al., 2014
Piska, Poland	Temperate	*P. sylvestris*	2002	2005, 2011, 2013	Managed forest	Free succession Salvage logging	Dobrowolska, 2015

* The geographical location of the site is taken from the information provided by the author(s) of each case study.

mounds or vegetative regrowth. First we therefore assessed if the case study included results that could help in answering our research questions. We divided results into three categories: (1). assessed quantitatively, (2). assessed qualitatively and (3). not assessed. To be classified as 'assessed quantitatively' the results should be reported in the results section whereas 'assessed qualitatively' only needed to be included in the discussion section.

For key research question #1 we continued the analysis by dividing each recovery process ('post storm regeneration', 'survivors', 'advanced regeneration' and 'vegetative regrowth') that was either assessed quantitatively or qualitatively, into three levels, (1). dominant presence, (2). co-dominant presence, and (3). minor or no presence. To be considered 'dominant', a process should be assessed quantitatively and its presence should be considerably higher than the presence of any of the other recovery processes, and the information should be noted by the authors. For a recovery process to be 'co-dominant' at least two processes should have a more pronounced influence on the recovery than any of the others. For a recovery process to be 'dominant' in a qualitative assessment, the author of the case study should note that this process dominated on the investigated site (Table 2).

For key research question #2, we assessed if the authors provided us with information on structures and processes that enhance or impede forest recovery, specifically we looked at pits and mounds, coarse woody debris, seed source proximity, any type of game interference e.g. browsing, bark stripping or fraying with antlers, competing ground vegetation and pests and diseases. We used a similar categorisation of importance, (1). important, (2). less important, and (3). not important. For an influencing process to be 'important' in a quantitative assessment, there should be a significant effect in the case study. For a process or structure to be 'less important', there should only be statistical indications noted by the authors of the case study, and to be 'not important', statistical testing should show insignificance without indications. Our assessment was designed so, because many of the results on influencing structures and processes were based on statistical analyses in the case studies. In the qualitative assessment of the importance by the case study authors, this should be substantiated by scientific literature (Table 2).

To assess key research question #3, eight sub-questions on the effect of salvage logging on forest recovery were assessed (Table 1). The conclusions of the authors of the case studies on the effect of salvage logging on forest recovery were tallied when they were specifically mentioned. Some authors registered multiple effects of salvage logging on forest recovery.

We analysed each case study for statistical results or qualitative discussions, which could answer our research questions. We further divided our results based on the location of the study site being within the boreal or the temperate biome to investigate if general patterns between the two biomes were different. Because so few studies were available we decided not to refine further than the distinction between boreal and temperate biomes. We based our classification of the biomes following the Global Forest Resources Assessment of the Food and Agriculture Organization (FRA, 2012), with an exception for the sites of the European case studies that FAO classifies as 'continental temperate', which we classify as boreal because of their tree composition. Due to the limited number of studies investigating the effect of salvage logging compared to non-logging in the boreal biome, this division was only done for key research questions #1 and #2.

As some of our categories relied on subjective judgement, especially for studies that included a qualitative assessment, all 34 case studies were assessed by two authors to reduce personal perceptions and misinterpretation of the results. When two authors disagreed, a third author analysis was performed (Table A1).

3. Results

Of the 34 reviewed case studies, 13 studies were conducted in the boreal biome (ten from Europe and three from eastern Canada) and 21 studies were conducted in the temperate biome (13 from Europe and eight from USA). The European studies were well distributed between forests from northern boreal Russia to temperate northern France (Table 2, Fig. 1). The North American studies were less well distributed due to the large geographical area and the low number of studies, although the studies covered 9 states and did not cluster (Table 2, Fig. 1).

3.1. Research efforts

All 34 case studies investigated forest recovery after windthrow (key research question 1) although the focus was on different processes. The main focus was on post storm regeneration, which was considered in all studies and was quantitatively assessed in 31 studies and qualitatively assessed in three studies (Table 3). Of the 34 studies, 25 studies included an assessment of the role of advanced regeneration, and of those, 19 assessed this quantitatively and six qualitatively (Table 3). Of all the reviewed case studies, 21 studies also focused on the influence of surviving trees. However, only 13 studies assessed this quantitatively whereas eight studies performed a qualitative assessment (Table 3). Finally, 10 studies included an assessment of vegetative regrowth, and only three of these made a quantitative assessment (Table 3).

Of the 34 studies, 31 included an assessment of one or more influencing processes on forest recovery (key research question 2). Most focused on soil disturbances, in which pits and mounds from uprooted trees were the most investigated structure. Of the 24 studies assessing effects of soil disturbance, 18 included a quantitative assessment (Table 3). There was also some focus, seven qualitatively and 16 quantitatively, on presence of coarse woody debris as a microsite for post storm regeneration and protection from browsing. Only two studies did a quantitative assessment and ten did a qualitative assessment of seed source proximity (Table 3). Game interference and competition from ground vegetation received some research focus, with 17 and 19 studies respectively. For both, around half made a quantitative assessment. Pests and diseases were only considered in six studies, of which only one did a quantitative assessment (Table 3).

18 studies included a quantitative assessment of recovery in salvage logged forests (key research question 3) compared to non-salvaged forests and none did a qualitative assessment (Table 3).

3.2. Importance of structures and processes for forest ecosystem recovery

3.2.1. Recovery processes (key research question 1)

Post storm regeneration was considered the dominant forest regeneration process in 16 studies, it was considered co-dominant in 15 studies, and only three studies found this to be a minor process (Table 4). Both crown expansion from surviving trees and the presence

Table 3
Research efforts (number of investigations) on recovery processes and influencing structures and processes. Te is temperate, Bo is boreal.

	Assessed quantitatively		Assessed qualitatively		Not assessed
	Total	*Te/Bo*	Total	*Te/Bo*	Total
Recovery processes (key research question 1)					
Post storm regeneration	31	*14/17*	3	*0/3*	0
Surviving trees	13	*6/7*	8	*4/4*	13
Advanced regeneration	19	*12/7*	6	*3/3*	9
Vegetative regrowth	3	*2/1*	7	*4/3*	24
Influencing structures and processes (key research question 2)					
Soil disturbance	18	*12/6*	6	*3/3*	10
Coarse woody debris	16	*11/5*	7	*4/3*	11
Seed source proximity	2	*1/1*	10	*6/4*	22
Game interference	8	*6/2*	9	*8/1*	17
Competing ground vegetation	11	*9/2*	8	*3/5*	15
Pests and diseases	1	*0/1*	5	*2/3*	28
Salvage logging	18	*14/4*	0	*0/0*	16

Table 4
Importance of recovery processes (key research question 1) and influencing structures and processes (research question 2). Te is temperate, Bo is boreal.

	Dominant		Co-dominant		Minor	
	Total	*Te/Bo*	Total	*Te/Bo*	Total	*Te/Bo*
Recovery processes (key research question 1)						
Post storm regeneration	16	*13/3*	15	*6/9*	3	*2/1*
Surviving trees	2	*2/0*	16	*7/9*	3	*2/1*
Advanced regeneration	7	*2/5*	15	*10/5*	3	*3/0*
Vegetative regrowth	4	*3/1*	1	*0/1*	5	*1/4*
	Important		**Less important**		**Not important**	
Influencing structures and processes (key research question 2)						
Soil disturbance	17	*10/7*	6	*4/2*	1	*1/0*
Coarse woody debris	14	*9/5*	9	*5/4*	0	*0/0*
Seed source proximity	9	*6/3*	3	*1/2*	0	*0/0*
Game interference	10	*10/0*	6	*4/2*	1	*0/1*
Competing ground vegetation	11	*8/3*	8	*4/4*	0	*0/0*
Pests and diseases	4	*1/3*	2	*1/1*	0	*0/0*

of advanced regeneration were in many studies considered co-dominant together with post storm regeneration. However, where advanced regeneration was considered a dominant regeneration process in seven cases, surviving trees were only considered dominant in two studies. Vegetative regrowth was only considered dominant in four studies and co-dominant in one study (Table 4). Post storm regeneration seemed to be the dominant recovery process in the temperate biome, whereas for the boreal biome, advanced regeneration was deemed the dominant regeneration process, but with post storm regeneration also being dominant in some cases. Advanced regeneration was however, also often considered co-dominant, together with post storm regeneration and surviving trees (Table 4).

3.2.2. Enhancing or impeding structures and processes (key research question 2)

Soil disturbance and seed source proximity were considered important processes positively influencing establishment of trees in most case studies both in the temperate and the boreal biome (Table 4). Tree establishment was also affected positively in many case studies by the presence of coarse woody debris, both in the boreal and temperate biome. Competition from ground vegetation and game interference negatively influenced tree establishment, with the latter being the most important negative effect in the temperate biome and the former having equal importance in both biomes (Table 4). Out of the six studies that assessed outbreaks of pest and diseases, four found this process important for forest recovery (Table 4).

3.2.3. Influence of salvage logging (key research question 3)

All 18 salvage logged forests recovered after windthrow and subsequent salvage logging. However, four studies found that the salvage logged forests had a slower recovery rate compared to non-salvaged forests. More than half of the studies found salvage logging to change the successional pathway by pushing forests towards an earlier successional stage, including longer periods dominated by pioneer species (Table 5). Four studies found that competition from ground vegetation slowed down recovery compared to the non-salvaged sites. Five studies found the same effect for game interference. Four of these were the same studies which reported that salvage logging following windthrow slowed down forest recovery (Table 5, Table A1).

4. Discussion

The reviewed studies showed the importance of several processes for forest recovery after windthrow e.g. post storm regeneration, soil disturbances and seed source proximity, despite large site heterogeneity. However, Kramer et al. (2014) found that advanced regeneration and the presence of coarse woody debris were less important, which may indicate that some processes are more dependent on site specific conditions, pre-storm successional stage or tree species and structural composition.

4.1. Tree regeneration processes

Post storm regeneration, i.e. germination from seed after the storm event, was found to be the most dominant forest regeneration process in the temperate biome, whereas in the boreal forest both advanced regeneration from existing seedlings and saplings and post storm regeneration were comparably important. *P. abies* and *P. mariana* dominated the forest in all but two cases situated in the boreal biome. In some cases they were accompanied by species of the *Abies* genus. Both these *Picea* and *Abies* species are known to make extensive use of advanced regeneration (Greene et al., 1999; Tjoelker et al., 2007), thus making advanced regeneration an important regeneration process in the boreal biome. The species diversity in the case studies from the temperate biome had a larger diversity and therefore regeneration strategies also varied more. This puts the post storm regeneration in the forefront as the most common forest regeneration strategy, with advanced regeneration being present especially when shade tolerant species such as *F. sylvatica* and *A. alba* were part of the tree species mix (Fischer et al., 2002; Nagel et al., 2006; Fidej et al., 2016). Although most boreal case studies investigated mainly *Picea* sp. which are capable of regenerating in limited light availability, studies that were dealing with large scale disturbances in this biome also considered post storm regeneration important (Kuuluvainen and Juntunen, 1998; Shorohova et al., 2008; Vodde et al., 2015). The reviewed cases showed that increased light availability after large scale disturbances may benefit pioneer species, such as *Betula* sp. and *P. sylvestris* after windthrow (Kuuluvainen and Kalmari, 2003; Shorohova et al., 2008; Vodde et al., 2010). The share of pioneer species versus late successional species was in many cases determined by spatial scale, frequency and type of disturbance (Ilisson et al., 2007). Specifically, Shorohova et al. (2008) show that the amount of *Betula* sp. in forests dominated by *Picea* sp. depends both on the frequency and spatial scale of the disturbance. Moreover, the amount of *Betula* in boreal forests dominated by *P. sylvestris* is not only dependent on the scale and frequency, but also on the type of disturbance (Kuuluvainen and Juntunen, 1998). Sites with frequent, large scale disturbances will therefore have a larger proportion of pioneer species and vice versa. Ulanova (2000) also proposes that the difference between regeneration processes in temperate and boreal biomes may not be an effect of the biome *per se* but rather an effect of the dominating tree species and spatial and temporal scales of disturbances.

Table 5
Influence of salvage logging on forest recovery (number of investigations having influence). 18 investigations have assessed influence of salvage logging on forest recovery (Table 3). Some investigations have reported more than one effect.

Prevents forest recovery	Retards forest recovery	No effect	Changes successional pathway	Increases competition from ground vegetation	Increases game induced damages	Enhances recovery due to soil scarification	No effect noted
0	4	0	12	4	5	1	1

Although surviving trees were only considered the dominant forest regeneration process in two of the reviewed studies (Dobrowolska, 2015), it was considered co-dominant in many of the other case studies. This may be because many reviewed studies did not observe surviving trees as agents of canopy closure, but merely that these were acting as suppliers of seeds or agents of favourable microclimate for new establishment (Nagel et al., 2006; Vodde et al., 2015). However, *F. sylvatica* has a large crown plasticity even during old age (Schröter et al., 2012; Pretzsch, 2014) and its capacity to close small to medium canopy gaps following single tree and small disturbances should not be underestimated.

Vegetative regrowth was only assessed quantitatively in the reviewed studies that dealt with species which are known to resprout from stumps and root suckers, e.g. *Populus tremuloides* (Man et al., 2013). Downed trees regenerating through branches taking over the apical dominance were only found in a few of the reviewed cases (Lang et al., 2009; Royo et al., 2016).

4.2. Structures and processes influencing forest recovery

In the reviewed studies microsites were considered important structures for post storm regeneration both in temperate and boreal forests. The result confirms many other studies showing that exposed mineral soil provides a 'window of opportunity' for the regeneration of trees and ground vegetation (Eriksson and Fröborg, 1996; Zackrisson et al., 1997; Hautala et al., 2008; Hekkala et al., 2014). The level of importance seemed to differ with the biome or tree species. Uprooting of trees creates pits and mounds with a gradient of soil and microclimatic conditions that allows several species to establish that need specific growing conditions that are not met in undisturbed forests (Ulanova, 2000). In the reviewed studies, pits and mounds were deemed to be important for the establishment of *P. abies* ranging from very important (Kuuluvainen and Kalmari, 2003; Kramer et al., 2014) to showing no difference in its preference between pits, mounds and undisturbed soil (Ilisson et al., 2007). In the boreal forests, early successional broadleaved species with light seeds such as *Betula* sp. find a niche in disturbed areas like windthrown forests (Kuuluvainen et al., 1993). In the temperate biome, the pattern seems more diffuse. In forests dominated by *F. sylvatica*, none of the reviewed studies specifically mentioned the availability of pits and mounds of vital importance for regeneration. Nagel et al. (2006) suggest that intermediate disturbance, likely aided by high stand stability in managed forests, is not sufficient for maintaining minor accompanying species such as *U. glabra* and *A. pseudoplatanus*, even in a situation of abundant pit and mound structures. However, in a longer time perspective pits and mounds may also be important for the tree species dominating European temperate forests (Šebková et al., 2012), indicating that the studies included here may have had too short of a time perspective to make firm conclusions.

The importance of coarse woody debris varied substantially. Where Kuuluvainen and Kalmari (2003) found that *P. abies* was more frequently regenerating on coarse woody debris than on undisturbed soil, Vodde et al. (2010) found no regenerating trees on coarse woody debris, giving a rather diffuse pattern on the role of coarse woody debris. The main difference between Vodde et al. (2010) and Kuuluvainen and Kalmari (2003) was, however, the timing of the studies after the windthrow. Where Vodde et al. (2010) focused on tree establishment right after windthrow, Kuuluvainen and Kalmari (2003) studied a relatively long term perspective, and consequently coarse woody debris at a more advanced decay stage, indicating that coarse woody debris needs to be at a late decay stage to be a suitable microsite for seedling establishment (Girard et al., 2014). This may explain why we found varying outcomes on the importance of coarse woody debris.

Proximity of seed sources was found to be an important structure for recovery of forests both in temperate and boreal forests although only few reviewed studies assessed this quantitatively. However, Jonasova et al. (2010) found that even for large scale disturbances, establishment of both pioneer and shade tolerant species took place after windthrow, despite long distances to seed sources. This may indicate that the soil seed bank is important, and thus that seed source proximity after a storm event is less important in such cases. Karlsson and Nilsson (2005) propose that tree establishment in forests with a large soil seed bank may therefore be more dependent on favourable microsites for germination such as soil scarification and microclimatic conditions. Seeds of different tree species use different dispersal mechanisms, e.g. wind dispersal versus heavy seed dropping directly under trees. Thus, seed source proximity intrinsically may not be determining how fast forests recover, but may rather dictate which trees colonise the windthrown areas.

For pests and diseases there were no clear differences between temperate and boreal biomes, and only one of the reviewed studies made a quantitative assessment (Khakimulina et al., 2016). Most likely the minor role of infestation by pests and diseases in the reviewed case studies is due to the scope of the studies having focus on forest regeneration processes and not on pests and diseases (Gardiner et al., 2013). Although bark beetle attacks subsequent to windthrow are often feared in commercial forests, especially if timber is left on site (Wermelinger et al., 1999), the reviewed case studies did not confirm, nor reject this (Jonasova et al., 2010; Kramer et al., 2014).

There was a clear indication that game interference was considered more important in temperate biomes than in boreal biomes. The minor effect of game interference in the boreal case studies may originate from a lower game density in boreal forests compared to temperate forests or a feeding habit which may target non-commercial species and is therefore less researched. However, one of the reviewed cases found that regeneration, especially for deciduous trees on clear-cut sites, was heavily influenced by deer browsing, especially by moose in boreal Europe (Gotmark and Kiffer, 2014). Thus, our result, that game interference was less important in the boreal region, may be a result of the windthrow studies being conducted in forest reserves (Table 2), where salvage logging may be less performed and downed trees can act as barriers for browsers (Ilisson et al., 2007; Jonasova et al., 2010; Kramer et al., 2014).

For competing ground cover vegetation the reviewed studies found, as for game interference, that it was considered more important where salvage logging had been undertaken, as crowns may shade out light demanding ground cover species such as grasses (Fidej et al., 2016). This may also explain why the reviewed studies found competition from ground vegetation to be more important in temperate case studies, as it was here the salvage logging was most intensively investigated.

4.3. Influence of salvage logging

Salvage logging often changed the successional pathway by pushing forests to earlier successional stages, with larger proportions of pioneer species with increased intensity of salvage logging (Ilisson et al., 2007). This indicates that two consecutive disturbances may have a cumulative effect on the recovery, as hypothesised by Paine et al. (1998). If cumulative severity of the compounded disturbances is sufficiently high, or where interactions between several disturbances occur, drastic change in forest structures, ecosystem processes and functions, and species composition may occur, without recovery to pre-disturbance conditions (Lindenmayer et al., 2004; Lindenmayer and Noss, 2006).

Contrarily, Kramer et al. (2014) found only a very limited effect of salvage logging and suggested that site factors had much more impact on forest recovery and successional pathways. Salvage logging was, however, most likely more carefully performed in the study of Kramer et al. (2014) than in applied forestry, and therefore the cumulative effects less severe. Additionally, Peterson and Leach (2008a) concluded that severity of neither windthrow nor salvage logging was sufficiently strong to pass a threshold, leading to irreversible changes in the forest ecosystem, although they recognise that more severe storms and large scale salvage logging, including land clearing after salvage logging, may lead to detrimental effects which induce a permanently changed state. When we weigh the outcomes of these two reviewed studies by Peterson and Leach (2008a) and Kramer et al. (2014), we see that changes in the successional pathway of the forest are likely to occur.

Most of the reviewed studies, however, only assessed the first decade after the disturbance and therefore did not give an indication of this being a permanent or just a temporary effect. Long term studies on recovery after windthrow and salvage logging are rare. The case studies which covered the longest time span with continued observation since the disturbance were 30 years (Pontailler et al., 1997) and 27 years (Fischer and Fischer, 2012). The latter study demonstrated that *Betula* sp. was much more abundant in a salvage logged site compared to an unmanaged site in a *P. abies* dominated forest. In another reviewed study by Fischer et al. (2002) the long term effect was modelled. They found that the effect will persist for many decades, but diminish with time. Some studies assessed a later stage of recovery after 40 years (Gotmark and Kiffer, 2014) or 60 years (Kuuluvainen and Kalmari, 2003), but these were not able to register the effect of subsequent windthrows on recovery. Only the case study by Pontailler et al. (1997) was able to assess the effect of several windthrows and their cumulative effect on forest succession (Pontailler et al., 1997). It seems a homogenisation in tree species composition towards shade tolerant species was taking place. However, even in this old forest reserve, the historical anthropogenic influence is not clearly distinguishable from the disturbance-recovery dynamics.

We assessed that the studies reporting a slower recovery on the salvaged sites due to competition from ground vegetation, were the same studies that reported that game interference slowed down the recovery. One of the reviewed studies relates the salvage logging directly to the game interference by concluding that the removal of crowns, undermined the protection for tree regeneration (Jonasova et al., 2010). Other reviewed studies have also reported that regenerating trees on salvage logged sites experienced more competition from ground vegetation (Fidej et al., 2016), especially along tracks from skidding machinery (Dodet et al., 2011).

Schütz et al. (2006) assess *P. abies* as particularly vulnerable to windthrow. However, the debris of windthrown *P. abies* plays a vital role in the species' regenerative cycle, regenerating mostly and best on its own decaying wood (Tjoelker et al., 2007). Salvage logging, without subsequent replanting may thus seriously inhibit the establishment of post storm regeneration.

Other reviewed studies found that salvage logging destroyed advanced regeneration and therefore also reduced the competitiveness of especially shade tolerant species (Nagel et al., 2006; Royo et al., 2016). Destruction of advanced regeneration and reduced protection from coarse woody debris against game interference and competition from ground vegetation may, in part, explain the change of successional pathway towards earlier successional stages of salvage logged sites.

4.4. Management implications

Our study gives insight into natural forest recovery processes and structures that may accelerate forest recovery and thus be of value for forest managers. Common for post storm regeneration and advanced regeneration is that it requires seed sources and favourable conditions for seed germination, favourable microclimate for establishment and protection from game and competing ground vegetation. Examples of measures that might be taken in this context are leaving a certain amount of deadwood logs and crowns for protection and microclimate and saving survivors which provide similar benefits and additionally function as a seed source. Since deadwood is important for forest biodiversity (Hekkala et al., 2014; Hekkala et al., 2016) the measure also provides a way to provide multiple forest benefits, a growing trend along with the increasing awareness of the non-wood benefits of forests (Haara et al., 2018).

The reviewed studies demonstrated that especially soil disturbances providing mineral soil were important for seed germination of early successional species immediately after the disturbance. Other authors such as Karlsson and Nilsson (2005) substantiated this by demonstrating a positive effect of soil scarification on seed germination. The effect of soil scarification is, however, also strongly influenced by available seed sources (Karlsson and Nilsson, 2005). Thus, stable trees of various tree species having the potential to survive storms and act as seed sources after storms can be incorporated into managed forests and potentially enhance natural recovery after windthrow and other ecosystem services (Pukkala, 2016). Secondly, we demonstrated that deadwood in later stages of decay was important for seed germination, especially for spruce species. As such, forest managers could incorporate more deadwood in the forests, hereby enhancing the regeneration potential, and additionally create habitats for biodiversity (Muller and Butler, 2010). For both coarse woody debris and pits and mounds, our results indicate windows of opportunity for their importance in the recovery process. For the first few years after the windthrow, pits and mounds seem to provide mineral soil, much as soil scarification does, offering germination options for early successional species. Deadwood primarily serves as a physical barrier against browsing and provides a more favourable micro-climate for germination (Rammig et al., 2007). When this stage has passed after a few years, it may take time for both pits, mounds and deadwood to start the second phase of their contribution to recovery. The results seem to indicate that for late-successional species, pits and mounds need to become more stable to function as a substrate, whereas deadwood needs to be in an advanced decay state. The aforementioned processes and structures are legacies of pre-storm forest conditions, most likely playing a key role for recovery and future forest ecosystem resilience (Jõgiste et al., 2017). There is limited knowledge on the integration of retaining disturbance legacies into forest management over extended periods in the forests development. A current societal interest in a more nature focussed forest management may provide opportunities for the long term integration of disturbance legacies into forest management. Stem only harvest leaving intact crowns uniformly distributed on the site after salvage logging, may reduce negative effects of game interference and competition from ground vegetation on new regenerating trees in the window of opportunity for seedling establishment directly after the disturbance. Given the nature of some legacies, their integration might still give rise to conflict with the optimal forest management for other ecosystem resources or economical gains.

We agree with Newton and Cantarello (2015) that one of the challenges faced by forest managers is to decide whether the result of their management is a target reference or ecosystem, or if a more open-ended outcome is acceptable. We will therefore not propose a one-size-fits-all management suggestion here, but generally argue for clearly defining the range of desired outcomes before embarking on management choices responding to forest disturbances. From our analyses, it is apparent that natural disturbances and salvage logging offer different opportunities for managing forest recovery. In the light of changing climatic conditions and increasing disturbances (Seidl et al., 2014; Müller et al., 2018), it stems that the more forest management deviates from the conditions shaped by natural processes of succession and disturbance, the more it might be forced to rely on building in measures of resistance.

In forests with limited accessibility, cost of salvage logging and replanting, may exceed revenue from extraction (Thorn et al., 2018). As such, passive restoration 'doing nothing' would be a viable option as we here demonstrated that forest recovery was retarded on salvage logged sites. Royo et al. (2016), specifically mention that particularly forests with shade tolerant species, which are dependent on advanced regeneration, often will be destroyed during harvest operations. The uncertainty in timber outcome encumbered with passive restoration, emphasises the need for a shift in management paradigm towards a stronger focus on resilience, where disturbances are regarded as an opportunity for managing of a range of different ecosystem services on a landscape scale (Seidl et al., 2016). If advanced regeneration is present and to be used for forest recovery after windthrow, care should be taken in salvaging the windthrown timber (Kramer et al., 2014). If accompanied by retaining legacies in the form of survivors, deadwood logs and crowns this may be a low cost measure to accelerate recovery in heavily damaged areas, while maintaining future harvest potential in the areas less affected by the windthrow.

Although passive restoration is encumbered with risks in relation to specific ecosystem services, time to recover and successional pathway of recovery, our study suggests that boreal and temperate forests are capable of recovering from both windthrow and salvage logging. The choice between active or passive restoration is one with important implications for forest economy, because of the costly actions of active restoration versus the lack of interventions and hence uncertainty encumbered with passive restoration. These outcomes indicate that passive restoration after forest disturbances may be a viable choice where both forest biodiversity and economy could profit, under the condition that the specifics of the forest are of lesser importance.

4.5. Method evaluation

By using a systematic quantitative literature review, we were able to provide a reproducible and reliable assessment of the current status on research investigating forest recovery after windthrow (Roy et al., 2012). Unfortunately, we were not able to systematically include the possibly important factor of repeated or interacting disturbances and the consequent presence of disturbance legacies, due to a lack of registered data on this score.

Using peer reviewed studies for reviews, *meta*-analyses or systematic quantitative reviews, always bears the risk of a bias where researchers and reviewers tend to favour publications including significant effects (Pickering and Byrne, 2014; Hedin et al., 2016). In our study this is especially the case for the comparison of forest recovery in unsalvaged sites versus salvage logged sites. The risk is that studies which do not find effects of salvage logging have not been published. Despite this potential error, the approach allowed recognitions of some general patterns in forest recovery after windthrow and the effects of salvage logging.

Appendix

(See Table A1)

4.6. Research needs

Given the large geographical space that our study covered, 34 studies is a relatively low number to show general patterns given the large site variability. Thus, in general there is need of more research on recovery of forests after windthrow. Moreover there were contradicting results for several of the structures and processes investigated in this study (Table 3). Structures and processes that have been insufficiently studied centre around the conditions under which coarse woody debris is favourable for seedling establishment, the importance of seed source proximity for different tree species and the interactions between salvage logging and both game interference and competing vegetation for seedling establishment and survival.

In the reviewed case studies, the time between the windthrow and the data collection only allows for the assessment of the early stages of the forest recovery process. The scarcity of case studies specifically assessing recovery in the context of successive windthrows or disturbances interacting with windthrow may also be tackled with long-term studies, paired in this case with a research design and approach that accounts for these effects. As we commonly think of successional stages in forests taking decades or even centuries, the lack of long term studies on forest dynamics is problematic. Conclusions on successional pathway and stipulations about later seral stages must therefore be made with caution. Suggestions to move this field forward would entail maintaining existing field experiments for repeated future research, using field data from time series to model predicted developments and combination with neighbouring fields which might have valuable knowledge to reconstruct previous conditions such as archaeology, pollen analysis and tree age determination. In order to become aware of the opportunities that natural forest recovery has to offer for the delivery of ecosystem services, a more intensive and long term study of forest succession in relation to disturbances is imperative.

It remains uncertain to which degree retaining disturbance legacies conflicts with forest management for other ecosystem services such as wood production, biodiversity, water regulation, carbon sequestration or recreation. An increased research effort should uncover how to practically fit these legacies of pre-storm conditions into current management practices. Additionally we might add that there is a lack of knowledge on how the different successional stages after forest recovery tailor to a set of ecosystem services. The recent *meta*-analysis by Thorn et al. (2018) on biodiversity after windthrow is a good addition in this sense. However, comparative studies of how different management types after windthrow affect especially timber quality, carbon sequestration and ground water quality are still lacking. As a final remark on future research needs, we wish to mention that from this study it becomes evident that a research effort leading to practical forest management suggestions in response to windthrow is lacking.

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Table A1
Each study is assessed with either one number (3) if topic is not assessed, or two numbers (X,X), where the first number refers to if the topic is assessed quantitatively (1) or assessed qualitatively (2). The second number refers to the answer to a specific research question (See Table 1) with the numbering in the top row refering to each sub research question (Table 1. The number can for forest recovery processes (Post storm regeneration, Surviving trees, Advanced regeneration, Vegetative regrowth) take the value 1-3 with 1, 2 and 3 referring to dominant, co-dominant or no/minor, respectively. For influencing structures and processes (Soil disturbance, Coarse woody debris, Seed source proximity, Game interference, Competing ground vegetation, Pests and diseases) the number can also be 1-3 referring to important, less important and not important. For salvage logging the second and following numbers can take the values 1-8, referring to: Prevents forest recovery, Retards forest recovery, Has no effect on forest recovery, Changes the successional pathway of forest recovery, Increases the competition from ground vegetation, Increases game interference to the recovering forest, Enhances tree regeneration due to soil scarification, No effect is noted (See Table 1). The temperate and boreal biomes are denoted t and b, respectively. The last column refers to the authors that performed the assessment of the specific case study.

REF. NO.	Refs.	(1a) Post storm regeneration	(1b) Surviving trees	(1c) Advanced regeneration	(1d) Vegetative regrowth	(2a) Soil disturbance	(2b) Coarse woody debris	(2c) Seed source proximity	(2d) Game interference	(2e) Competing ground vegetation	(2f) Pests and diseases	(3) Salvage logging	Biome	Assessed by
1	Wolf et al.	2,3	2,1	3	3	3	3	3	3	3	3	3	t	Taeroe/de Koning
2	Bigler and Wolf	2,2	2,2	2,2	2,1	1,1	1,2	2,1	3	3	3	3	t	Taeroe/de Koning
3	Bolte et al.	2,2	2,2	3	3	3	3	3	3	3	2,1	3	b	Taeroe/de Koning
4	Götmark and Kiffer	1,2	1,2	3	2,3	2,2	3	2,1	2,2	2,1	3	1,8	b	Taeroe/de Koning
5	Leemans	1,2	1,2	1,1	3	2,1	2,2	1,1	3	2,2	2,2	3	b	Taeroe/de Koning
6	Kuuluvainen and Kalmari	1,1	2,2	2,2	3	1,1	1,1	3	3	2,1	3	3	b	Taeroe/de Koning
7	Kuuluvainen and Juntunen	1,1	1,2	3	2,3	1,1	1,1	3	3	2,1	3	3	b	Taeroe/de Koning
8	Ilisson et al.	1,1	2,2	2,2	3	1,1	2,2	3	3	3	3	1,4,6	b	Taeroe/de Koning
9	Vodde et al. A	1,2	2,2	1,2	2,3	1,1	1,2	2,2	1,2	3	3	3	b	Taeroe/de Koning
10	Vodde et al. B	1,2	1,2	2,2	3	1,1	2,1	2,1	1,3	2,2	3	3	b	Taeroe/de Koning
11	Khakimulina et al.	1,2	1,2	1,1	3	3	3	3	3	3	1,1	3	b	Taeroe/de Koning
12	Shorohova et al.	1,3	1,2	1,1	3	3	3	3	3	3	3	3	b	Taeroe/de Koning
13	Harmer and Morgan	1,1	1,3	1,3	2,1	3	3	3	1,2	3	3	1,4	t	Taeroe/de Koning
14	Harmer et al.	1,1	3	3	3	3	3	2,1	2,2	1,1	3	1,4	t	Taeroe/de Koning
15	Dodet et al.	1,1	3	3	3	1,2	1,1	3	3	1,1	3	1,2,5	t	Taeroe/de Koning
16	Pontallier et al.	1,1	1,2	1,2	1,2	1,2	1,2	1,1	2,1	2,1	3	3	t	Taeroe/de Koning
17	Kramer et al.	1,1	3	1,3	3	1,2	1,2	3	1,1	1,1	2,2	1,5,6,7	t	Taeroe/de Koning
18	Jonášová et al.	1,2	2,2	1,2	3	1,1	1,1	2,2	2,2	1,2	2,1	1,2,5,6	t	Taeroe/de Koning
19	Fidej et al.	1,2	3	2,2	3	2,1	1,2	2,1	1,2	1,2	3	1,2,5,6	t	Taeroe/de Koning/ Raulund-Rasmussen
20	Fischer et al.	1,2	3	1,1	3	1,1	3	3	3	1,2	3	1,4	t	Taeroe/de Koning
21	Fischer and Fischer	1,1	3	1,3	3	2,1	2,1	3	3	3	3	1,4	t	Taeroe/de Koning

(continued on next page)

Table A1 (*continued*)

REF. NO.	Refs.	(1a) Post storm regeneration	(1b) Surviving trees	(1c) Advanced regeneration	(1d) Vegetative regrowth	(2a) Soil disturbance	(2b) Coarse woody debris	(2c) Seed source proximity	(2d) Game interference	(2e) Competing ground vegetation	(2f) Pests and diseases	(3) Salvage logging	Biome	Assessed by
22	Nagel et al.	1,3	2,2	1,1	3	1,3	1,1	3	1,1	2,2	3	3	t	Taeroe/de Koning
23	Peterson and Pickett	1,1	3	2,2	3	1,1	3	3	1,1	1,1	3	3	t	Taeroe/de Koning
24	Peterson and Leach A	1,1	3	3	3	1,1	1,1	3	2,1	1,1	3	1,4	t	Taeroe/de Koning
25	Holzmueller et al.	1,1	1,2	3	3	3	3	3	3	3	3	3	t	Taeroe/de Koning
26	Allen et al.	1,1	1,2	3	3	2,1	2,1	3	3	3	3	3	t	Taeroe/de Koning
27	Waldron et al.	1,2	3	1,1	3	1,1	1,2	3	3	1,2	3	1,4	b	Taeroe/de Koning
28	Lang et al.	1,2	1,3	1,2	1,1	1,2	2,2	2,1	2,1	3	3	1,4	t	Taeroe/de Koning
29	Royo et al.	1,1	3	1,2	2,3	1,1	1,1	3	2,1	1,1	3	1,4	t	Taeroe/de Koning
30	Man et al.	1,2	1,3	1,2	1,1	3	1,1	2,2	3	1,2	3	1,4	b	Taeroe/de Koning
31	Petersson and Leach B	1,1	3	1,2	3	1,1	1,1	3	2,1	2,1	3	1,4	t	Taeroe/de Koning
32	White et al.	1,1	3	1,2	3	3	3	3	2,1	3	3	1,4	t	Taeroe/de Koning
33	Girard et al.	1,2	3	1,1	3	2,2	1,1	3	3	3	2,1	3	b	Taeroe/de Koning
34	Dobrowolska	1,2	1,1	1,2	2,3	3	2,1	2,1	1,1	3	3	1,2,6	t	Taeroe/de Koning

References

Allen, M.S., Thapa, V., Arevalo, J.R., Palmer, M.W., 2012. Windstorm damage and forest recovery: accelerated succession, stand structure, and spatial pattern over 25 years in two Minnesota forests. Plant Ecol. 213, 1833–1842.

Anyomi, K.A., Mitchell, S.J., Ruel, J.C., 2016. Windthrow modelling in old-growth and multi-layered boreal forests. Ecol. Model 327, 105–114.

Birch, J.C., Newton, A.C., Aquino, C.A., Cantarello, E., Echeverria, C., Kitzberger, T., Schiappacasse, I., Garavito, N.T., 2010. Cost-effectiveness of dryland forest restoration evaluated by spatial analysis of ecosystem services. P. Natl. Acad. Sci. USA 107, 21925–21930.

Bigler, J., Wolf, A., 2007. Structural impact of gale damage on Suserup Skov, a near-natural temperate deciduous forest in Denmark. Ecol. Bull. 52, 69–80.

Bolte, A., Hilbrig, L., Grundmann, B., Kampf, F., Brunet, J., Roloff, A., 2010. Climate change impacts on stand structure and competitive interactions in a southern Swedish spruce-beech forest. Eur. J. For. Res. 129, 261–276.

Buma, B., Wessman, C.A., 2011. Disturbance interactions can impact resilience mechanisms of forests. Ecosphere 2, art64.

Coutts, M.P., Nielsen, C.C.N., Nicoll, B.C., 1999. The development of symmetry, rigidity and anchorage in the structural root system of conifers. Plant Soil 217, 1–15.

Davis, M.A., Wrage, K.J., Reich, P.B., 1998. Competition between tree seedlings and herbaceous vegetation: support for a theory of resource supply and demand. J. Ecol. 86, 652–661.

Dobrowolska, D., 2015. Forest regeneration in northeastern Poland following a catastrophic blowdow n. Can. J. For. Res. 45, 1172–1182.

Dodet, M., Collet, C., Frochot, H., Wehrlen, L., 2011. Tree regeneration and plant species diversity responses to vegetation control following a major windthrow in mixed broadleaved stands. Eur. J. Forest Res. 130, 41–53.

Engelmark, O., Hofgaard, A., Arnborg, T., 1998. Successional trends 219 years after fire in an old Pinus sylvestris stand in northern Sweden. J. Veg. Sci. 9, 583–592.

Eriksson, O., Fröborg, H., 1996. Windows of opportunity for recruitment in long-lived clonal plants: experimental studies of seedling establishment in Vaccinium shrubs. Can. J. Bot. 74, 1369–1374.

Fidej, G., Rozman, A., Nagel, T.A., Dakskobler, I., Diaci, J., 2016. Influence of salvage logging on forest recovery following intermediate severity canopy disturbances in mixed beech dominated forests of Slovenia. Iforest 9, 430–436.

Fischer, A., Fischer, H.S., 2012. Individual-based analysis of tree establishment and forest stand development within 25 years after wind throw. Eur. J. Forest Res. 131, 493–501.

Fischer, A., Lindner, M., Abs, C., Lasch, P., 2002. Vegetation dynamics in central European forest ecosystems (near-natural as well as managed) after storm events. Folia Geobot. 37, 17–32.

Fra, F., 2012. Global ecological zones for FAO forest reporting: 2010 Update. Rome, Italy, FAO.

Gardiner, B., Berry, P., Moulia, B., 2016. Review: Wind impacts on plant growth, mechanics and damage. Plant Sci. 245, 94–118.

Gardiner, B., Schuck, A.R.T., Schelhaas, M.J., Orazio, C., Blennow, K., Nicoll, B. (Eds.), 2013. Living with storm damage to forests. European Forest Institute, Joensuu, pp. 1–132.

Girard, F., De Grandpre, L., Ruel, J.C., 2014. Partial windthrow as a driving process of forest dynamics in old-growth boreal forests. Can. J. For. Res. 44, 1165–1176.

Gotmark, F., Kiffer, C., 2014. Regeneration of oaks (Quercus robur/Q-petraea) and three other tree species during long-term succession after catastrophic disturbance (windthrow). Plant Ecol. 215, 1067–1080.

Greene, D.F., Zasada, J.C., Sirois, L., Kneeshaw, D., Morin, H., Charron, I., Simard, M.-J., 1999. A review of the regeneration dynamics of North American boreal forest tree species. Can. J. For. Res. 29, 824–839.

Holzmueller, E.J., Gibson, D.J., Suchecki, P.F., 2012. Accelerated succession following an intense wind storm in an oak-dominated forest. For. Ecol. Manage. 279, 141–146.

Haara, A., Pykäläinen, J., Tolvanen, A., Kurttila, M., 2018. Use of interactive data visualization in multi-objective forest planning. J. Environ. Manage. 210, 71–86.

Harmer, M., Tucker, N., Nickerson, R., 2004. Natural regeneration in storm damaged woods – 1987 storm sites revisited. Quart. J. For. 98 (3).

Harmer, R., Morgan, G., 2009. Storm damage and the conversion of conifer plantations to native broadleaved woodland. For. Ecol. Manag. 258, 879–886.

Hautala, H., Tolvanen, A., Nuortila, C., 2008. Recovery of pristine boreal forest floor community after selective removal of understorey, ground and humus layers. Plant Ecol. 194, 273–282.

Hedin, R.J., Umberham, B.A., Detweiler, B.N., Kollmorgen, L., Vassar, M., 2016. Publication bias and nonreporting found in majority of systematic reviews and meta-analyses in anesthesiology journals. Anesth. Analg. 123, 1018–1025.

Hekkala, A.-M., Tarvainen, O., Tolvanen, A., 2014. Dynamics of understory vegetation after restoration of natural characteristics in the boreal forests in Finland. For. Ecol. Manag. 330, 55–66.

Hekkala, A.-M., Ahtikoski, A., Päätalo, M.-L., Tarvainen, O., Siipilehto, J., Tolvanen, A., 2016. Restoring volume, diversity and continuity of deadwood in boreal forests. Biodivers. Conserv. 25, 1107–1132.

Ilisson, T., Koster, K., Vodde, F., Jogiste, K., 2007. Regeneration development 4–5 years after a storm in Norway spruce dominated forests, Estonia. For. Ecol. Manage. 250, 17–24.

Jactel, H., Bauhus, J., Boberg, J., et al., 2017. Tree diversity drives forest stand resistance to natural disturbances. Curr. Forest. Rep. 3, 20.

Jõgiste, K., Korjus, H., Stanturf, J.A., Frelich, L.E., Baders, E., Donis, J., Jansons, A., Kangur, A., Köster, K., Laarmann, D., 2017. Hemiboreal forest: natural disturbances and the importance of ecosystem legacies to management. Ecosphere 8.

Jonasova, M., Vavrova, E., Cudlin, P., 2010. Western Carpathian mountain spruce forest after a windthrow: natural regeneration in cleared and uncleared areas. For. Ecol. Manag. 259, 1127–1134.

Karlsson, M., Nilsson, U., 2005. The effects of scarification and shelterwood treatments on naturally regenerated seedlings in southern Sweden. For. Ecol. Manag. 205, 183–197.

Khakimulina, T., Fraver, S., Drobyshev, I., 2016. Mixed-severity natural disturbance regime dominates in an old-growth Norway spruce forest of northwest Russia. J. Veg. Sci. 27, 400–413.

Kramer, K., Brang, P., Bachofen, H., Bugmann, H., Wohlgemuth, T., 2014. Site factors are more important than salvage logging for tree regeneration after wind disturbance in Central European forests. For. Ecol. Manag. 331, 116–128.

Kuijper, D.P.J., Cromsigt, J.P.G.M., Churski, M., Adam, B., Jedrzejewska, B., Jedrzejewski, W., 2009. Do ungulates preferentially feed in forest gaps in European temperate forest? For. Ecol. Manag. 258, 1528–1535.

Kuuluvainen, T., Juntunen, P., 1998. Seedling establishment in relation to microhabitat variation in a windthrow gap in a boreal Pinus sylvestris forest. J. Veg. Sci. 9, 551–562.

Kuuluvainen, T., Kalmari, R., 2003. Regeneration microsites of Picea abies seedlings in a windthrow area of a boreal old-growth forest in southern Finland. Ann. Bot. Fenn. 40, 401–413.

Kuuluvainen, T., Hokkanen, T.J., Järvinen, E., Pukkala, T., 1993. Factors related to seedling growth in a boreal Scots pine stand: a spatial analysis of a vegetation–soil system. Can. J. For. Res. 23, 2101–2109.

Lang, K.D., Schulte, L.A., Guntenspergen, G.R., 2009. Windthrow and salvage logging in an old-growth hemlock-northern hardwoods forest. For. Ecol. Manag. 259, 56–64.

Leemans, R., 1991. Canopy gaps and establishment patterns of spruce (*Picea-Abies* (L) Karst) in 2 old-growth coniferous forests in Central Sweden. Vegetatio 93, 157–165.

Leverkus, A.B., Gustafsson, L., Benayas, J.M.R., Castro, J., 2015. Does post-disturbance salvage logging affect the provision of ecosystem services? a systematic review protocol. Environ. Evid. 4, 16.

Lindenmayer, D.B., Noss, R.F., 2006. Salvage logging, ecosystem processes, and biodiversity conservation. Conserv. Biol. 20, 949–958.

Lindenmayer, D.B., Foster, D.R., Franklin, J.F., Hunter, M.L., Noss, R.F., Schmiegelow, F.A., Perry, D., 2004. Ecology – salvage harvesting policies after natural disturbance. Science 303 1303-1303.

Man, R.Z., Chen, H.Y.H., Schafer, A., 2013. Salvage logging and forest renewal affect early aspen stand structure after catastrophic wind. For. Ecol. Manag. 308, 1–8.

Mason, W.L., 2002. Are irregular stands more windfirm? Forestry 75, 347–355.

Michalova, Z., Morrissey, R.C., Wohlgemuth, T., Bace, R., Fleischer, P., Svoboda, M., 2017. Salvage-logging after windstorm leads to structural and functional homogenization of understory layer and delayed spruce tree recovery in Tatra Mts., Slovakia. Forests 8.

Muller, J., Butler, R., 2010. A review of habitat thresholds for dead wood: a baseline for management recommendations in European forests. Eur. J. For. Res. 129, 981–992.

Müller, J., Noss, R.F., Thorn, S., Bässler, C., Leverkus, A.B., Lindenmayer, D., 2018. Increasing disturbance demands new policies to conserve intact forest. Conserv. Lett. e12449.

Myking, T., Bøhler, F., Austrheim, G., Solberg, E.J., 2011. Life history strategies of aspen (Populus tremula L.) and browsing effects: a literature review. For.: Int. J. For. Res. 84, 61–71.

Nagel, T.A., Svoboda, M., Diaci, J., 2006. Regeneration patterns after intermediate wind disturbance in an old-growth Fagus-Abies forest in southeastern Slovenia. For. Ecol. Manag. 226, 268–278.

Nakagawa, M., Kurahashi, A., Hogetsu, T., 2003. The regeneration characteristics of Picea jezoensis and Abies sachalinensis on cut stumps in the sub-boreal forests of Hokkaido Tokyo university forest. For. Ecol. Manag. 180, 353–359.

Newton, A.C., Cantarello, E., 2015. Restoration of forest resilience: an achievable goal? New For. 46, 645–668.

Paine, R.T., Tegner, M.J., Johnson, E.A., 1998. Compounded perturbations yield ecological surprises. Ecosystems 1, 535–545.

Peterson, C.J., Leach, A.D., 2008b. Salvage logging after windthrow alters microsite diversity, abundance and environment, but not vegetation. Forestry 81, 361–376.

Peterson, C.J., Leach, A.D., 2008a. Limited salvage logging effects on forest regeneration after moderate-severity windthrow. Ecol. Appl. 18, 407–420.

Peterson, C.J., Pickett, S.T.A., 1995. Forest reorganization – a case-study in an old-growth forest catastrophic blowdown. Ecology 76, 763–774.

Peterson, C.J., Pickett, S.T.A., 2000. Patch type influences on regeneration in a western Pennsylvania, USA, catastrophic windthrow. Oikos 90, 489–500.

Pickering, C., Byrne, J., 2014. The benefits of publishing systematic quantitative literature reviews for PhD candidates and other early-career researchers. High Educ. Res. Dev. 33, 534–548.

Pontailler, J.Y., Faille, A., Lemee, G., 1997. Storms drive successional dynamics in natural forests: a case study in Fontainebleau forest (France). For. Ecol. Manag. 98, 1–15.

Pretzsch, H., 2014. Canopy space filling and tree crown morphology in mixed-species stands compared with monocultures. For. Ecol. Manag. 327, 251–264.

Provendier, D., Balandier, P., 2008. Compared effects of competition by grasses (Graminoids) and broom (Cytisus scoparius) on growth and functional traits of beech saplings (Fagus sylvatica). Ann. For. Sci. 65 510-510.

Pukkala, T., 2016. Which type of forest management provides most ecosystem services? For. Ecosyst. 3, 9.

Rammig, A., Fahse, L., Bebi, P., Bugmann, H., 2007. Wind disturbance in mountain forests: simulating the impact of management strategies, seed supply, and ungulate browsing on forest succession. For. Ecol. Manag. 242, 142–154.

Roy, S., Byrne, J., Pickering, C., 2012. A systematic quantitative review of urban tree benefits, costs, and assessment methods across cities in different climatic zones. Urban Urban Gree 11, 351–363.

Royo, A.A., Peterson, C.J., Stanovick, J.S., Carson, W.P., 2016. Evaluating the ecological impacts of salvage logging: can natural and anthropogenic disturbances promote coexistence? Ecology 97, 1566–1582.
Samonil, P., Kral, K., Hort, L., 2010. The role of tree uprooting in soil formation: a critical literature review. Geoderma 157, 65–79.
Schröter, M., Härdtle, W., von Oheimb, G., 2012. Crown plasticity and neighborhood interactions of European beech (Fagus sylvatica L.) in an old-growth forest. Eur. J. For. Res. 131, 787–798.
Schütz, J.-P., Götz, M., Schmid, W., Mandallaz, D., 2006. Vulnerability of spruce (Picea abies) and beech (Fagus sylvatica) forest stands to storms and consequences for silviculture. Eur. J. For. Res. 125, 291–302.
Šebková, B., Šamonil, P., Valtera, M., Adam, D., Janík, D., 2012. Interaction between tree species populations and windthrow dynamics in natural beech-dominated forest, Czech Republic. For. Ecol. Manage. 280, 9–19.
Seidl, R., Rammer, W., Spies, T.A., 2014. Disturbance legacies increase the resilience of forest ecosystem structure, composition, and functioning. Ecol. Appl. 24, 2063–2077.
Seidl, R., Spies, T.A., Peterson, D.L., Stephens, S.L., Hicke, J.A., 2016. Searching for resilience: addressing the impacts of changing disturbance regimes on forest ecosystem services. J. Appl. Ecol. 53, 120–129.
Senf, C., Seidl, R., 2018. Natural disturbances are spatially diverse but temporally synchronized across temperate forest landscapes in Europe. Glob. Change Biol. 24, 1201–1211.
Shorohova, E., Fedorchuk, V., Kuznetsova, M., Shvedova, O., 2008. Wind-induced successional changes in pristine boreal Picea abies forest stands: evidence from long-term permanent plot records. Forestry 81, 335–359.
Thorn, S., Bässler, C., Brandl, R., Burton, P.J., Cahall, R., Campbell, J.L., Castro, J., Choi, C.Y., Cobb, T., Donato, D.C., 2018. Impacts of salvage logging on biodiversity: a meta-analysis. J. Appl. Ecol. 55, 279–289.
Tjoelker, M.G., Boratynski, A., Bugala, W., 2007. Biology and ecology of Norway spruce. Springer Science & Business Media.
Ulanova, N.G., 2000. The effects of windthrow on forests at different spatial scales: a review. For. Ecol. Manag. 135, 155–167.
Vodde, F., Jogiste, K., Gruson, L., Ilisson, T., Koster, K., Stanturf, J., 2010. Regeneration in windthrow areas in hemiboreal forests: the influence of microsite on the height growths of different tree species. J. For. Res.-Jpn. 15, 55–64.
Vodde, F., Jogiste, K., Kubota, Y., Kuuluvainen, T., Koster, K., Lukjanova, A., Metslaid, M., Yoshida, T., 2011. The influence of storm-induced microsites to tree regeneration patterns in boreal and hemiboreal forest. J. For. Res.-Jpn. 16, 155–167.
Vodde, F., Jogiste, K., Engelhart, J., Frelich, L.E., Moser, W.K., Sims, A., Metslaid, M., 2015. Impact of wind-induced microsites and disturbance severity on tree regeneration patterns: results from the first post-storm decade. For. Ecol. Manag. 348, 174–185.
Waldron, K., Rue, J.C., Gauthier, S., 2013. Forest structural attributes after windthrow and consequences of salvage logging. For. Ecol. Manag. 289, 28–37.
Wermelinger, B., Obrist, M., Duelli, P., Forster, B., 1999. Development of the bark beetle (Scolytidae) fauna in windthrow areas in Switzerland. Mitteilungen-Schweizerische Entomologische Gesellschaft 72, 209–220.
White, S.D., Hart, J.L., Cox, L.E., Schweitzer, C.J., 2014. Woody regeneration in a Southern Appalachian Quercus stand following wind disturbance and salvage logging. Castanea 79, 223–236.
Wolf, A., Moller, P.F., Bradshaw, R.H.W., Bigler, J., 2004. Storm damage and long-term mortality in a semi-natural, temperate deciduous forest. For. Ecol. Manag. 188, 197–210.
Zackrisson, O., Nilsson, M.-C., Dahlberg, A., Jäderlund, A., 1997. Interference mechanisms in conifer-Ericaceae-feathermoss communities. Oikos 209–220.